Azad Ellafi

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EDUCATION

Boston University, Boston, MA. *May 2026*
Bachelor of Arts in Computer Science

EXPERIENCE

Yale University, CS Research Intern. New Haven, CT. *Jun 2023 – Aug 2023*

- Supported development of a digital archive for historical CS publications, facilitating improved access for researchers and educators.
- Designed a searchable database to catalog and retrieve academic data efficiently.
- Engaged in archival research of historical computer science publications.
- Implemented cleaning workflows to improve the accuracy and consistency of archived publication metadata.

City of Boston & BU Spark, Air Quality Analysis. Boston, MA. *Sep 2023 – Dec 2023*

- Developed a comprehensive data-driven report on transportation infrastructure's impact on air quality across 15+ Boston neighborhoods, drawing on public health datasets spanning 5+ years, for the City of Boston.
- Conducted geospatial and statistical analyses, identifying significant correlations between traffic density and PM2.5 levels across income brackets.
- Created 10+ interactive visualizations used in presentations to city planners and decision-makers.
- Automated portions of the data processing pipeline to streamline analysis and reduce manual error in recurring SCRUM reports.

PROJECTS

Stack-Based Language Interpreter, Programming Language Theory. *Boston University*

- Designed and implemented a fully functional interpreter in OCaml for a custom stack-based programming language as the capstone project for a Programming Language Theory course.
- Language supported a core instruction set including arithmetic operations (push, add, sub, mul, div), stack manipulation (pop, dup, swap), and conditional branching (if/else blocks).

SKILLS & AWARDS

Programming Languages: C, Python, Java, Julia, OCaml, Haskell, Nix

Databases & Tools: Git, KVM, Kubernetes, VMware, Docker, MySQL, Redis, SQL, noSQL

Operating Systems & Scripting: Linux, Bash, PowerShell

Concepts: Data Structures, Data Analysis, Machine Learning, Artificial Intelligence, Geospatial Analysis, Algorithmic Design, Data Pipeline Automation

Capstone Award Winner – 2022

Awarded top distinction (2022) for capstone "On the Non-Consensual Usage of Private Consumer Data." Conducted a multidisciplinary study of data collection across the internet, using major incident case studies to expose gaps between GDPR/CCPA and technical harvesting practices. Proposed a tiered consent framework and engineering mitigations with practical recommendations for developers and policymakers.